

The birth–death–catastrophe process: definition and main results

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Chapter 3 of AA Thesis — University of Leeds
Draft compiled August 9, 2026

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1 Introduction

We aim to understand the stochastics of those infections which exploit a lytic mechanism in their initial pathogenesis: infectious agents that use intracellular environments to their proliferative advantage. The importance of stochasticity here can hardly be overstated. During the first handful of replicative cycles there is an exaggerated chance that an infection dies out before it gets started, and a deterministic treatment risks glossing over this perilous beginning, its variables venturing smoothly between one bacterium and none. To grasp *why* *Y. pestis*, and other pathogens like it, do what they do, we must account for the interplay of chance and fortune in the smallest populations. It is for this reason that we have selected the birth–death–catastrophe process as our tool of inquiry.

Study of the birth–death–catastrophe (BDC) process and its dynamics forms a central focus of this thesis. Here we present a series of results and analyses associated with its behaviour. As a continuous-time, discrete-state Markov chain, the BDC process found its origin in a paper by Brockwell, Gani and Resnick published in 1982 [1]. During the late 1970s and early 1980s, attempts were being made to describe unpredictable population collapses using stochastic process theory:

“There has recently been a great deal of interest in the development and analysis of stochastic models for the growth of populations which are subject to catastrophes due either to death or large-scale emigrations” [1].

Prior to that work, models had largely incorporated deterministic growth interspersed by stochastic reductions — sudden drops, the sizes and times of which were defined in distribution (see, for example, the models of Pakes and collaborators from the late 1970s). These suffered from the peculiarities of modelling discrete populations with continuous random variables, and so attempts were made to employ a step process instead: a continuous-time Markov chain in a discrete state space.

In fact, the model presented in that 1982 paper is a generalisation of what we refer to as the BDC process. Not only did it also include immigration events — single steps up occurring as a Poisson process with constant rate — but the catastrophic steps down in population count were sized as random i.i.d. variables, with a few basic distributions investigated. In our case, the catastrophe event reduces the population directly to 0, or to a special post-catastrophe state H ; in either case causing fixation.

Our interest in the birth–death–catastrophe process stems from within-host infection modelling. Like others before us, we are interested in the pathogenic behaviour of intracellular replication and bursting. Our agent of focus, *Y. pestis*, finds its way into host macrophages and replicates there, before ultimately lysing the cell so that the population of infectious bacteria can traverse the extracellular matrix and continue its exploitative fight for survival. “Catastrophe” here names the cell-bursting event, which becomes increasingly likely as the intracellular population expands. Carruthers and collaborators worked on the same model with focus on the agent *Francisella tularensis* [2], and similar work has been directed towards the pathogenesis of salmonella. The mathematical backbone of these models rests on the classical theory of birth-and-death processes and their killed variants [3, 4, 5].

Our particular interest is the potential importance of this replication and bursting strategy, along with other behaviours of *Y. pestis*, for an advantage in early pathogenesis — getting a foothold when numbers are few, before a serious immune response arrives. Those questions return in Chapters 4a and 4b, where the single-cell theory developed here is completed and then embedded in a population model; for the purposes of the present chapter, we take the birth–death–catastrophe process as an abstract model, unbound by any specific application. We will present a number of mathematical

results which quantify the stochastics of its growth and fixation over time.

Further interest

Interest in the dynamics of catastrophic population decline is nothing new. Towards the end of the 18th century, Thomas Malthus published his fears of impending societal collapse. He worried that a geometrically growing census would be less and less supported by only arithmetic development in food production, ultimately resulting in mass famine. In the relevant literature, such population reductions by external factors are termed “positive checks”, as opposed to the “preventative checks” by which individuals have fewer children to account for resource scarcity.

In 1755, just 11 years before Thomas Malthus was born, Lisbon experienced a substantial earthquake that devastated the city. Voltaire includes a graphic scene of this in his allegorical novel *Candide*. Perhaps catastrophe was on the mind of the age, and maybe the anxiety was justified. France, after all, was on the brink of revolution:

“War had created such social systems of unprecedented complexity that they exceeded the capacities of their internal regulatory mechanisms to orchestrate the behaviours of their parts. When, as in France, administrative chaos became too great and the government collapsed — an event easily measured by the financial crisis of 1788 — society exploded.” [6]

This quote is from a paper which attempts to apply Prigogine’s theory of dissipative structures to the evolution of culture and societal norms through revolutions. It is perhaps ambitious, and lacking in empirical merit, but it has a certain appeal. Similarly, we are curious about the prospect that the BDC process, or a variant of it, can be taken in allegory to illustrate a general behaviour observed in dissipative structures present across multiple dynamic phenomena.

We have also been interested in the role of catastrophic events in triggering evolutionary transitions. A notable example is the KT boundary, where 65 million years ago much of life on earth was wiped out by an asteroid, setting the stage for a mass proliferation and diversification of mammals.

It has also been postulated that the catastrophes of plague in Europe and their aftermath contributed to advancements in technology and culture — most recently by James Belich, who argues that previous estimates for the scale of population decline sit below the real figure, claiming the initial wave saw a death of at least 50% of the European population in the decade following 1345. With such a sudden elimination of so many people, labour was in reduced supply. It has been hypothesised that a subsequent necessity to do more work with fewer people encouraged a number of mechanical inventions; these, along with higher wages, might have contributed to the improved prosperity and the cultural advancements seen in the renaissance and the enlightenment periods. It is even said that the Lisbon earthquake and its cleanup spurred advancements in geographical techniques: damage to buildings was diligently recorded by locality, providing new light on the concept of an epicentre.

These evolutionary echoes of catastrophe will be touched on again when, in Chapter 4b, we compare the bursting strategy against its alternatives. For now, let us take a step back, and then forward, into the mathematics of the birth–death–catastrophe process.

2 Reference values

Before the process is defined and its results derived, we collect in one place the symbols, the key quantities, and the parameter combinations that will be used throughout this chapter and the two

that follow it. Everything listed here is stated again, with derivation, in the body of the chapter; this page is the reference card.

Key parameters

λ : birth rate (per replicative unit)

μ : death rate (per replicative unit)

δ : catastrophe rate (per replicative unit)

Defining quantities

$\mathbf{X}_t$: process state, the intracellular count

$\mathbf{X}_t = 0$: internal extinction state (absorbing)

$\mathbf{X}_t = H$: catastrophe (rupture) state (absorbing)

$\mathbf{W}_t$: released count (0 before rupture, $\mathbf{X}_{\tau-}$ thereafter)

τ : catastrophe time

$\mathcal{K}$: burst size, the value of $\mathbf{X}_t$ immediately before rupture

Measures of stochasticity

$I(t)$: $\Pr\{\mathbf{X}_t \neq H\}$, non-catastrophe probability

$D(t)$: $p_0(t) = \Pr\{\mathbf{X}_t = 0\}$, probability of internal extinction (no burst)

$I_{\text{fix}}(t)$: $\Pr\{\mathbf{X}_t \notin \{0, H\}\} = I(t) - D(t)$, non-fixation probability (neither catastrophe nor death)

$J(t)$: $\mathbb{E}(\mathbf{X}_t)$, expected process state

$K(t)$: $\mathbb{E}(\mathbf{X}_t^2)$, second moment of the process state

$V(t)$: $\mathbb{E}(\mathbf{W}_t)$, expected released count

$p_n(t)$: $\Pr\{\mathbf{X}_t = n\}$, state probability distribution

P_k : $\Pr\{\mathcal{K} = k\}$, burst size distribution

$\varphi(t)$: $\delta J(t)$, probability density of the burst time τ (defective when $\mu > 0$)

Parameter combinations

$$\eta = \frac{\lambda + \mu + \delta}{2\lambda}$$

$$a = \eta + \sqrt{\eta^2 - \frac{\mu}{\lambda}} \quad (b < 1 < a)$$

$$b = \eta - \sqrt{\eta^2 - \frac{\mu}{\lambda}}$$

$$A, B, \theta = a - 1, \quad 1 - b, \quad \lambda(a - b)$$

$$\kappa = 1 + \frac{\delta}{2\lambda}$$

$$w = e^{\theta t} \text{ (the recurring exponential of the closed forms)}$$

The constants a and b are the two roots of the quadratic $\lambda x^2 - (\lambda + \mu + \delta)x + \mu = 0$, and A , B , θ measure distances from the unstable root 1. Their elementary identities, used constantly in what follows, are

$$ab = \frac{\mu}{\lambda}, \quad a + b = \frac{\lambda + \mu + \delta}{\lambda}, \quad AB = \frac{\delta}{\lambda}, \quad A + B = a - b, \quad \theta = \lambda(a - b), \quad (1)$$

each of which is immediate from Vieta's formulas together with the definitions $A = a - 1$ and $B = 1 - b$; they are re-derived in Section 5.

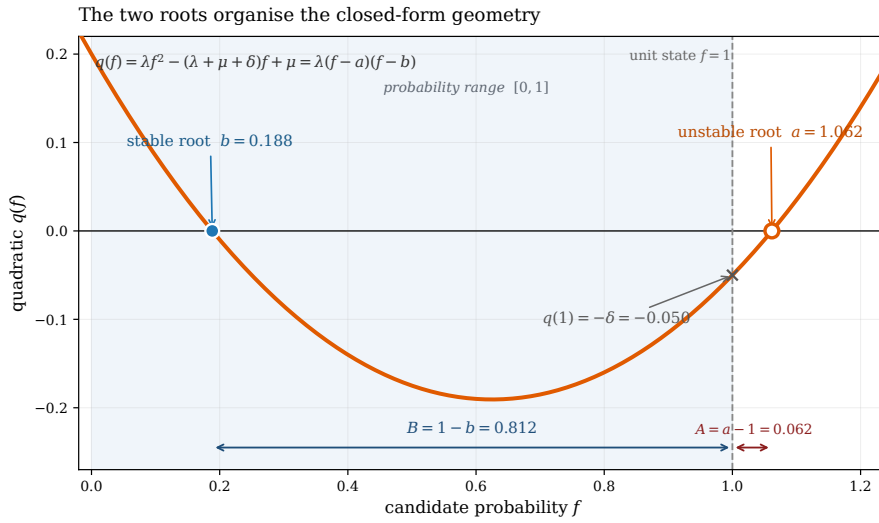


Figure 1: The characteristic quadratic for $\lambda = 1$, $\mu = 0.2$, and $\delta = 0.05$. Its roots are $b = 0.188 < 1 < a = 1.062$; the distances $B = 1 - b$ and $A = a - 1$ set the geometric ratios and late-time constants used throughout the chapter. Filled and open markers distinguish the stable and unstable roots of the Riccati flow, respectively.

Key ODEs

$$\frac{dI}{dt} = \mu - (\lambda + \mu + \delta)I + \lambda I^2 = \lambda(I - a)(I - b)$$

(Riccati equation for the non-rupture probability)

$$\frac{dI}{dt} = -\delta J$$

(forward equation for the rupture state)

$$\frac{dJ}{dt} = (\lambda - \mu)J - \delta K$$

(expectation of $\mathbf{X}_t$)

$$\frac{dV}{dt} = \delta K$$

(expectation of $\mathbf{W}_t$)

These four differential equations, together with the initial conditions $I(0) = J(0) = K(0) = 1$ and $V(0) = 0$, generate every closed form derived in this chapter.

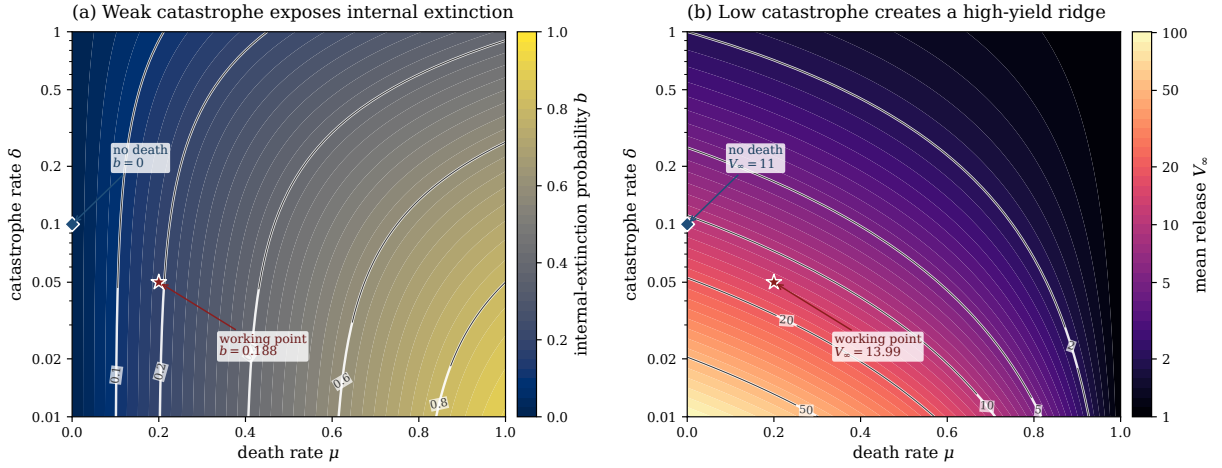


Figure 2: Rate landscapes at fixed $\lambda = 1$, shown for $\mu \in [0, 1]$ and $\delta \in [0.01, 1]$ on a logarithmic δ axis. (a) The lower root b is the probability of internal extinction before release. (b) The mean release V_∞ forms a sharp high-yield ridge as catastrophe becomes rare. The star marks the chapter's working point $(\mu, \delta) = (0.2, 0.05)$, where $b = 0.188$ and $V_\infty = 13.99$; the diamond marks the no-death point $(0, 0.1)$, where $b = 0$ and $V_\infty = 11$. Contours provide a colour-independent reading of both landscapes.

3 Process definition

Framed in the context of an intracellular infection which ultimately causes the host cell to burst, our chosen version of the birth–death–catastrophe model is constructed in the following way. $\mathbf{X}_t$ is a stochastic process of discrete state space, evaluated in continuous time. It satisfies the Markov property with transition probabilities dependent only on its current state:

$$\begin{aligned}
 \Pr \{ \mathbf{X}_{t+\Delta t} - \mathbf{X}_t = 1 \} &= \lambda \mathbf{X}_t \cdot \Delta t && \text{(birth)} \\
 \Pr \{ \mathbf{X}_{t+\Delta t} - \mathbf{X}_t = -1 \} &= \mu \mathbf{X}_t \cdot \Delta t && \text{(death)} \\
 \Pr \{ \mathbf{X}_{t+\Delta t} - \mathbf{X}_t = 0 \} &= 1 - (\lambda + \mu + \delta) \mathbf{X}_t \cdot \Delta t && \text{(no event)} \\
 \Pr \{ \mathbf{X}_{t+\Delta t} = H \} &= \delta \mathbf{X}_t \cdot \Delta t && \text{(catastrophe)}
 \end{aligned}$$

The count $\mathbf{X}_t$ is a number of replicative units, which in our application represent infectious particles. With rate λ , a given one of these units will undergo division, a birth event; with rate μ , a death event. And with rate δ , a given particle will trigger the catastrophe event. With this staged as rupture of the infected macrophage, all $\mathbf{X}_t$ of the virions or bacteria are released into the extracellular matrix.

In a duration Δt , the probability of a birth occurring is $\lambda \Delta t \cdot \mathbf{X}_t$: the likelihood of a single unit dividing, multiplied by the number of units $\mathbf{X}_t$. The model is evaluated in the limit $\Delta t \rightarrow 0$. This is why the above transition probabilities account for only the occurrence of single events; the likelihood that two or more transitions occur in Δt time is proportional to $o(\Delta t)$, and thus vanishes in the limit.

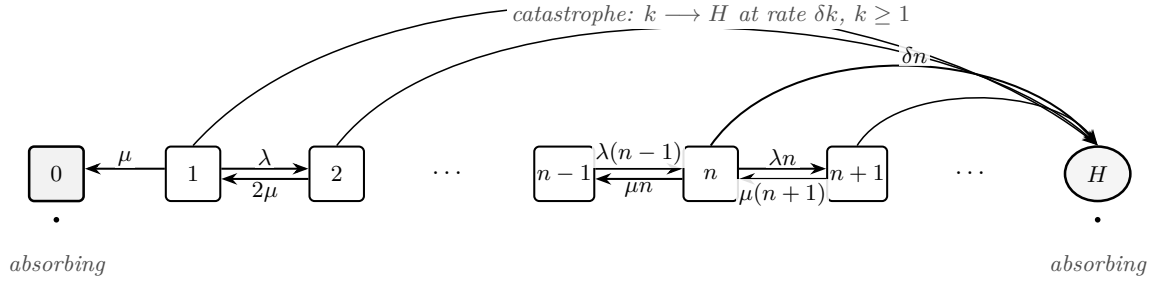


Figure 3: Transition diagram of the birth–death–catastrophe process. From a non-absorbing state n , births occur at rate λn , deaths at rate μn , and catastrophe moves the process to H at rate δn . States 0 and H are absorbing.

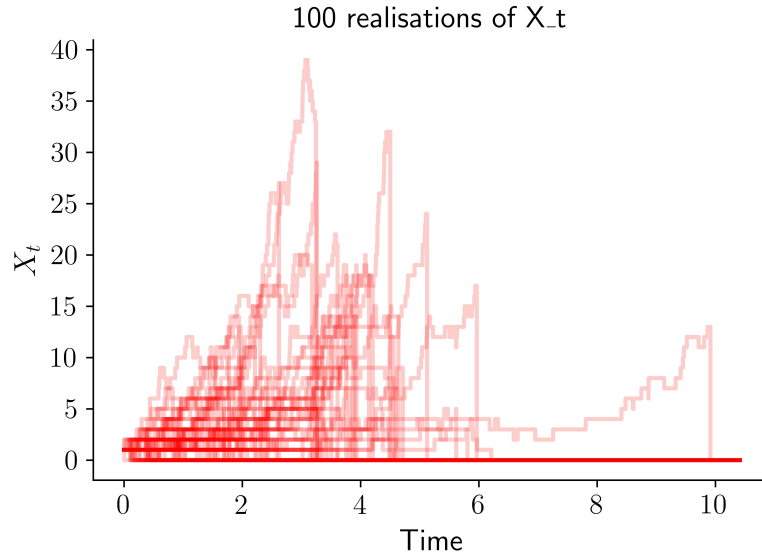


Figure 4: 100 realisations of $\mathbf{X}_t$ generated using the Gillespie algorithm. $\lambda = 1$, $\mu = 0$, $\delta = 0.1$.

Sample paths of the process are shown in Figure 4: realisations rise and fall stochastically, some collapsing to the absorbing state 0, others terminating abruptly at catastrophe.

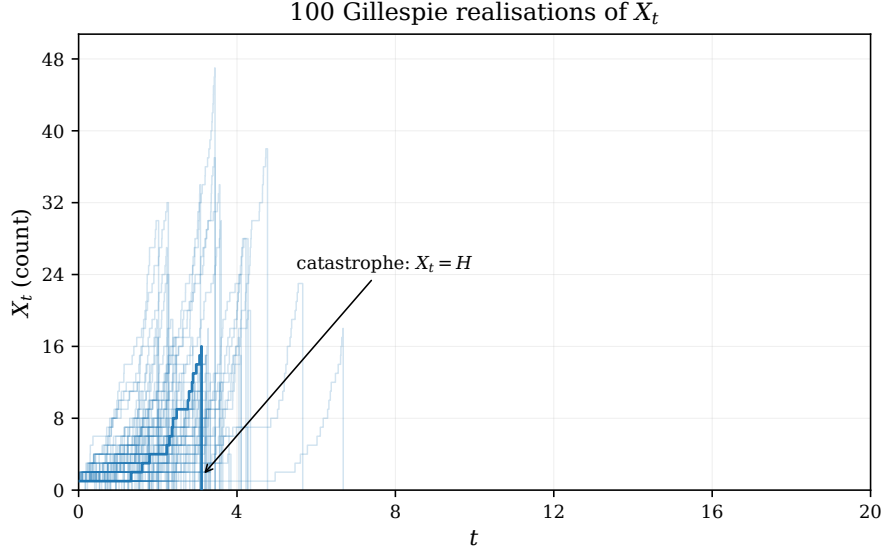


Figure 5: One hundred Gillespie realisations of X_t , re-rendered on $0 \leq t \leq 20$. The terminal step to the plotting baseline marks entry into the absorbing catastrophe state H , rather than the count state 0. $\lambda = 1$, $\mu = 0$, $\delta = 0.1$, and $X_0 = 1$.

A second ensemble, now resolved by terminal fate, exposes the structure hidden by an undifferentiated bundle of sample paths. In particular, the ensemble mean need not resemble any one of the realisations from which it is formed.

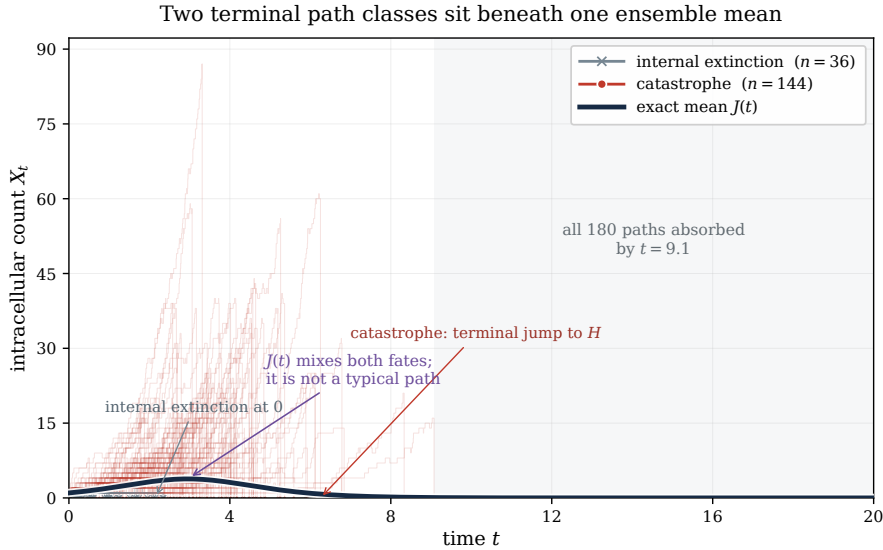


Figure 6: Terminal-fate decomposition of 180 exact Gillespie realisations of X_t . Blue-grey paths ending in crosses reach internal extinction at 0; red paths ending in filled circles enter the separate catastrophe state H . The bold curve is the exact mean $J(t) = \mathbb{E}[X_t]$. Because both absorbed classes subsequently contribute zero, this mean mixes structurally different histories rather than describing a typical path. Here $\lambda = 1$, $\mu = 0.2$, $\delta = 0.05$, $X_0 = 1$, and the fixed simulation seed is 3505.

3.1 Choice of rupture state

An alternative definition lists the “catastrophe state” as 0 instead of a separate state H , and it would make some intuitive sense to arrange the process like this. After all, following rupture of an infected cell, there are subsequently 0 replicative units within it — but then, the cell also no longer exists. Ultimately, whether we ascribe the catastrophe state to 0 or H is of no material significance to the dynamics: the burst times and sizes observed are unchanged. But a distinction is created in some of the formulas. These will be mentioned as we continue; for our purposes, the catastrophe transition is taken to the isolated state H .

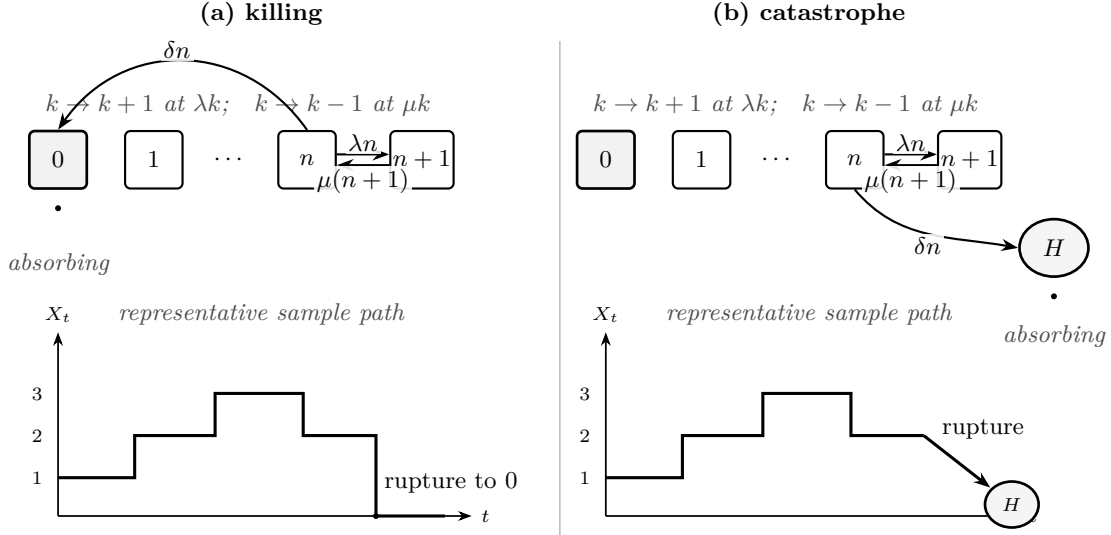


Figure 7: Killing and catastrophe conventions for rupture. Under the killing convention, a transition at rate δn returns the process from state n to the absorbing state 0. Under the catastrophe convention used here, the same transition ends at the separate absorbing state H ; the birth and death rates remain λn and μn , respectively.

3.2 Post-catastrophe count

In the context we are considering, the catastrophe signifies the rupture of an infected cell, and following this release some count of infectious units enters the extracellular medium. This post-rupture count, assuming no further dynamics, is fixed by the final value of $\mathbf{X}_t$ prior to catastrophe, and it remains at that value for all future time. We refer to this post-rupture count as $\mathbf{W}_t$, itself a stochastic process: $\mathbf{W}_t = 0$ for $t < \tau$, and $\mathbf{W}_t = \mathbf{X}_{\tau-}$ thereafter for $t \geq \tau$, where τ is the time of cell rupture. The process $\mathbf{W}_t$ is included in the definition as follows, to describe the full stochastic process $(\mathbf{X}_t, \mathbf{W}_t)$:

$$\begin{aligned}
 \Pr \{ \Delta \mathbf{X}_t = 1, \Delta \mathbf{W}_t = 0 \} &= \lambda \mathbf{X}_t \cdot \Delta t && \text{(birth)} \\
 \Pr \{ \Delta \mathbf{X}_t = -1, \Delta \mathbf{W}_t = 0 \} &= \mu \mathbf{X}_t \cdot \Delta t && \text{(death)} \\
 \Pr \{ \Delta \mathbf{X}_t = 0, \Delta \mathbf{W}_t = 0 \} &= 1 - (\lambda + \mu + \delta) \mathbf{X}_t \cdot \Delta t && \text{(no event)} \\
 \Pr \{ \mathbf{X}_{t+\Delta t} = H, \mathbf{W}_{t+\Delta t} = \mathbf{X}_t \} &= \delta \mathbf{X}_t \cdot \Delta t && \text{(catastrophe)}
 \end{aligned}$$

(using the shorthand $\Delta \mathbf{X}_t = \mathbf{X}_{t+\Delta t} - \mathbf{X}_t$).

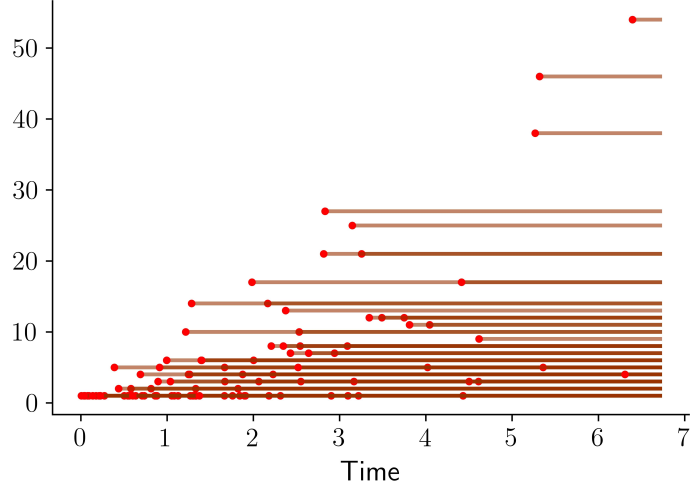


Figure 8: 100 realisations of $\mathbf{W}_t$ generated using the Gillespie algorithm. $\lambda = 1$, $\mu = 0$, $\delta = 0.1$.

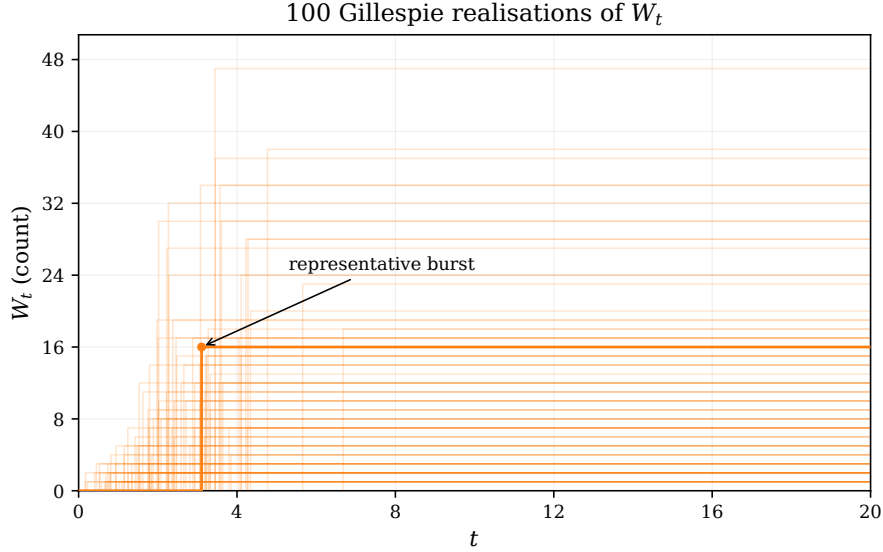


Figure 9: One hundred corresponding Gillespie realisations of W_t , re-rendered on $0 \leq t \leq 20$. Each path remains at zero until rupture, then holds at the released count $X_{\tau-}$; the arrow marks one representative burst. $\lambda = 1$, $\mu = 0$, $\delta = 0.1$, and $X_0 = 1$.

Figure 8 shows the corresponding released-count realisations: each path holds at zero until its cell ruptures, then jumps to the intracellular load accumulated at that instant and stays there. The pair $(\mathbf{X}_t, \mathbf{W}_t)$ is the object this chapter dissects, and the two components will be studied through their expectations, variances, and eventually their full distributions.

4 Analytical quantities

To understand the dynamical properties of the BDC process, there are a handful of quantities on which the analysis focuses. We introduce each in turn — the non-catastrophe probability, the

non-fixation probability, the process mean, and the expected released count — stating the closed forms first and deriving them in Section 5. Throughout, the process starts from a single replicative unit, $\mathbf{X}_0 = 1$, unless stated otherwise.

4.1 $I(t)$, the non-catastrophe probability

$$I(t) = \Pr \{ \mathbf{X}_t \neq H \mid \mathbf{X}_0 = 1 \}$$

Given that the rupture state H is accessible from every non-zero state of the Markov process, eventual catastrophe is inevitable provided the process does not first fixate by depletion to 0. Either death or catastrophe is therefore sure to happen at late time, and $I(t)$ is the probability that the process has not experienced catastrophe prior to time t .

Theorem 4.1. *The non-catastrophe probability is*

$$I(t) = \frac{aB + bAw}{B + Aw}, \quad (2)$$

where $w = e^{\theta t}$ with $\theta = \lambda(a - b)$, and

$$a = \eta + \sqrt{\eta^2 - \frac{\mu}{\lambda}}, \quad b = \eta - \sqrt{\eta^2 - \frac{\mu}{\lambda}}, \quad \eta = \frac{\lambda + \mu + \delta}{2\lambda},$$

are the two roots, $b < 1 < a$, of $\lambda x^2 - (\lambda + \mu + \delta)x + \mu = 0$, while $A = a - 1$ and $B = 1 - b$.

The constants a and b carry the probabilistic content of the formula. When $\mu > 0$, death of the process is possible, and $I(t)$ tends down to the positive constant value b , the likelihood of eventual process death:

$$\lim_{t \rightarrow \infty} I(t) = \lim_{t \rightarrow \infty} \frac{aB + bAw}{B + Aw} = \frac{bA}{A} = b.$$

When $\mu = 0$, death is impossible and $b = 0$: catastrophe strikes every realisation eventually, and $I(t)$ decays to 0.

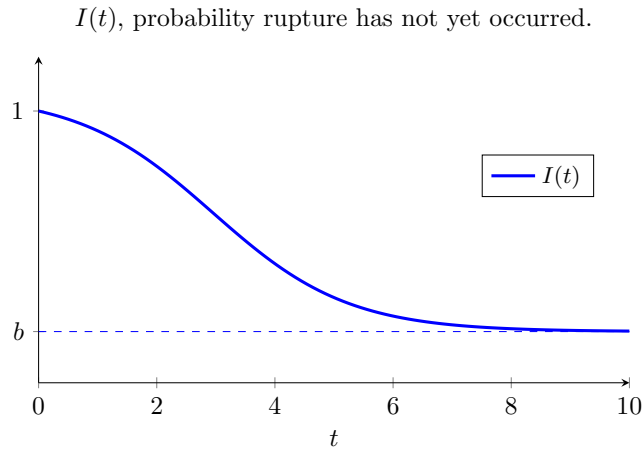


Figure 10: $I(t)$ plotted against time. It converges to b , the probability that the process dies out internally before rupture. $\lambda = 1$, $\mu = 0.2$, $\delta = 0.05$.

4.2 $I_{\text{fix}}(t)$, the non-fixation probability

$$I_{\text{fix}}(t) = \Pr \{ \mathbf{X}_t \notin \{0, H\} \mid \mathbf{X}_0 = 1 \}$$

We can also consider the probability that at time t the process has neither died out nor experienced catastrophe — that it remains “productive”. This is the non-fixation probability

$$I_{\text{fix}}(t) = \Pr \{ \mathbf{X}_t \notin \{0, H\} \mid \mathbf{X}_0 = 1 \},$$

which declines asymptotically to 0, signifying the eventual certainty of fixation one way or the other. Writing $D(t) = p_0(t)$ for the probability that the process has died out internally by time t , we have, since the absorbing states 0 and H exhaust the ways of leaving the productive states,

$$I_{\text{fix}}(t) = I(t) - D(t).$$

That decomposition pays off immediately: both I and D have closed forms (Theorems 4.1 and 4.2), and their difference simplifies.

Theorem 4.2. *The internal-extinction probability is*

$$D(t) = p_0(t) = \frac{ab(w-1)}{aw-b}, \quad (3)$$

and the non-fixation probability is

$$I_{\text{fix}}(t) = I(t) - D(t) = \frac{(a-b)^2 w}{(B+Aw)(aw-b)}. \quad (4)$$

The limits of (4) behave as the definition demands: $I_{\text{fix}}(0) = 1$ (the process begins productive), $I_{\text{fix}}(\infty) = 0$ (fixation is certain), and the initial slope is $I'_{\text{fix}}(0) = -(\mu + \delta)$: at the first instant, a productive realisation is lost either to death at rate μ or to catastrophe at rate δ . Notice also the collapse when $\mu = 0$: then $b = 0$ and $D(t) = 0$ at every t , so $I_{\text{fix}} = I$, and the two quantities of this section and the last coincide. When $\mu > 0$ they are genuinely distinct, and keeping them apart is what allows the quasi-stationary analysis of Chapter 4a to condition on continued productivity rather than mere survival of rupture. Figure 11 shows $I_{\text{fix}}(t)$ for the same parameters as Figure 10.

$\hat{I}(t)$, probability of non fixation.

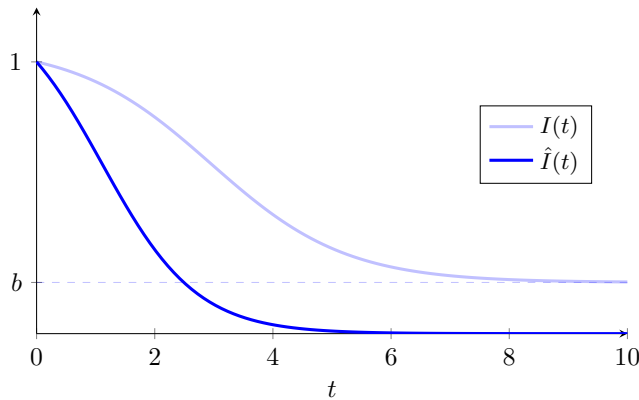


Figure 11: $I_{\text{fix}}(t)$ plotted against time. $\lambda = 1$, $\mu = 0.2$, $\delta = 0.05$.

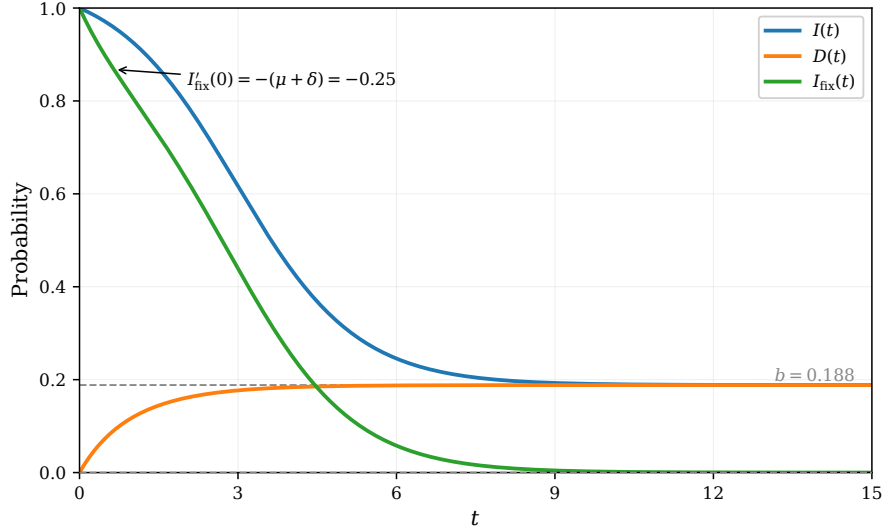


Figure 12: Non-catastrophe probability $I(t)$, internal-extinction probability $D(t)$, and non-fixation probability $I_{\text{fix}}(t) = I(t) - D(t)$. For $\lambda = 1$, $\mu = 0.2$, and $\delta = 0.05$, both $I(t)$ and $D(t)$ approach b , while $I_{\text{fix}}(t)$ tends to zero with initial slope $I'_{\text{fix}}(0) = -(\mu + \delta) = -0.25$.

4.3 Multiple founders

Suppose the process begins with k initial particles, $\mathbf{X}_0 = k$. Each replicative unit gives rise to its own family of descendants, all behaving independently of one another, so probabilities for the whole population factor into per-family probabilities (the branching property). The likelihood that any one of those k families will *not* have led to catastrophe by time t is $I(t)$; hence the probability that none of the k families has caused the mass emigration is

$$I_k(t) = \Pr \{ \mathbf{X}_t \neq H \mid \mathbf{X}_0 = k \} = [I(t)]^k.$$

Likewise, internal extinction of the whole population requires every family to die out, giving $D_k(t) = D(t)^k$, and for non-fixation more generally,

$$I_{\text{fix},k}(t) = \Pr \{ \mathbf{X}_t \notin \{0, H\} \mid \mathbf{X}_0 = k \} = I(t)^k - D(t)^k.$$

The tempting simplification $I_{\text{fix},k} = I_{\text{fix}}^k$ is false when $\mu > 0$: non-fixation of the whole population fails as soon as *any one* family fixates, whether by death or by rupture, and the two routes must be separated exactly as above. (When $\mu = 0$, $D = 0$ and the two expressions do coincide.) The multi-founder quantities return in Chapter 4a, where they are derived from the probability generating functions and used to describe cells infected by several particles at once.

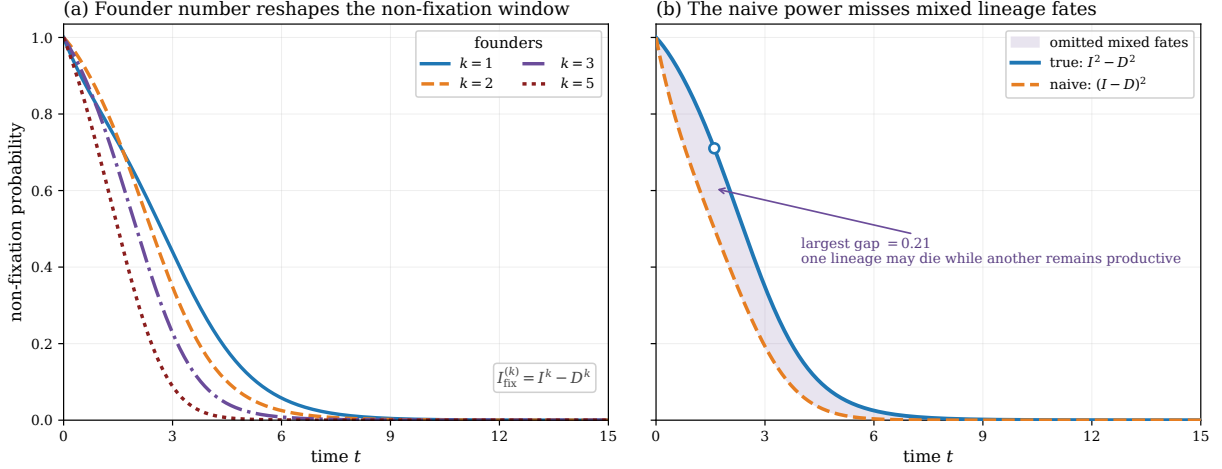


Figure 13: Multi-founder non-fixation. (a) The exact probabilities $I_{\text{fix},k}(t) = I(t)^k - D(t)^k$ for $k = 1, 2, 3, 5$. (b) For two founders, the naive power $I_{\text{fix}}(t)^2 = [I(t) - D(t)]^2$ lies below the exact result because it omits mixed configurations in which one lineage has died while another remains productive; the shaded band is precisely that missing probability. Parameters $\lambda = 1$, $\mu = 0.2$, and $\delta = 0.05$.

4.4 $J(t)$, the replicative unit mean

$$J(t) = \mathbb{E}(\mathbf{X}_t \mid \mathbf{X}_0 = 1)$$

A given realisation of the process will rise and fall stochastically in steps for some time before ending up stuck either at 0 or, after catastrophe, at H . Following each of these fixation outcomes, the contribution of $\mathbf{X}_t$ to the mean $J(t)$ is 0. So, supposing we observe 100 realisations, if at time $t = 20$ all of them have either died or undergone catastrophe, their sample mean would be 0 ($J_{\text{sample}}(20) = 0$). From the initial condition $\mathbf{X}_0 = 1$, the process mean rises while realisations tend to increase, then falls as the likelihood of fixation grows asymptotically certain.

$J(t) = \mathbb{E}(X_t)$, the expected number of replicators.

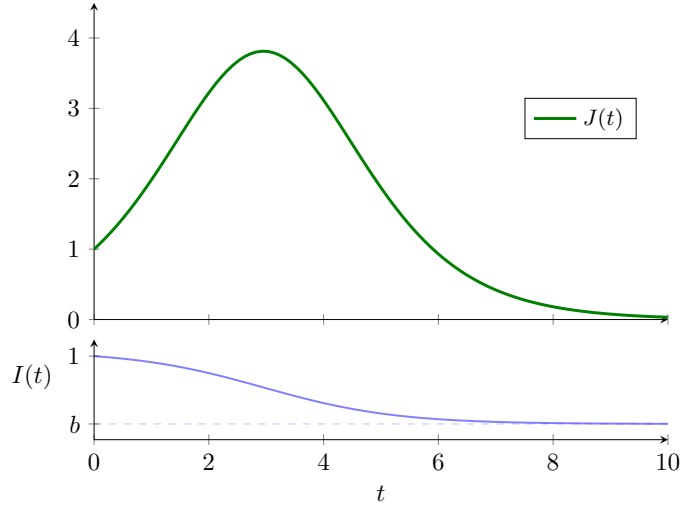


Figure 14: $J(t)$ plotted against time. $\lambda = 1$, $\mu = 0.2$, $\delta = 0.05$.

Theorem 4.3. *The expected process state is*

$$J(t) = \frac{(a-b)^2 w}{(B + Aw)^2} = -\frac{\lambda}{\delta} (I - a)(I - b), \quad (5)$$

obtained by differentiating $I(t)$ and multiplying by $-\delta^{-1}$, using the relation $I' = -\delta J$ established in Section 5.

The second form of (5), written purely in terms of I , is the one that will do most of the work later: it expresses the mean load as a function of the non-catastrophe probability alone.

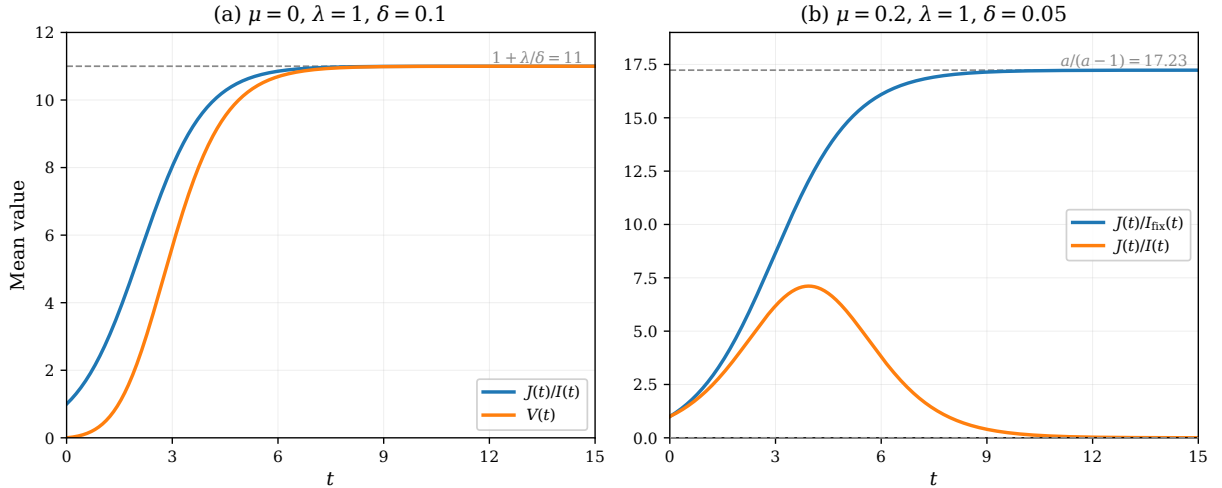


Figure 15: Conditional-mean convergence. (a) For $\mu = 0$, $\lambda = 1$, and $\delta = 0.1$, $J(t)/I(t)$ and $V(t)$ both tend to $1 + \lambda/\delta = 11$. (b) For $\lambda = 1$, $\mu = 0.2$, and $\delta = 0.05$, $J(t)/I_{\text{fix}}(t)$ tends to the quasi-stationary mean $a/(a-1)$, while $J(t)/I(t)$ tends to zero.

4.5 $V(t)$, the expected number of released units

$$V(t) = \mathbb{E}(\mathbf{W}_t \mid \mathbf{X}_0 = 1)$$

Over time, the process will either decline to 0, or transition to the catastrophe state H , in which case $\mathbf{W}_t$ is set at the final value of $\mathbf{X}_t$ before rupture. $V(t)$, the expected post-rupture count, incorporates all situations, whether $\mathbf{W}_t$ is zero or not.

Theorem 4.4. *The expected released count is*

$$V(t) = (1 - I) \left[1 + \frac{\lambda}{\delta} (1 - I) \right] = \frac{(\lambda - \mu)}{\delta} [1 - I(t)] - J(t) + 1, \quad (6)$$

and its late-time value is

$$V_\infty = \lim_{t \rightarrow \infty} V(t) = \frac{aB}{A} = \frac{\lambda - \mu}{\delta} (1 - b) + 1, \quad (7)$$

as $I \rightarrow b$ and $J \rightarrow 0$. When $\mu = 0$, this reduces to $V_\infty = 1 + \lambda/\delta$.

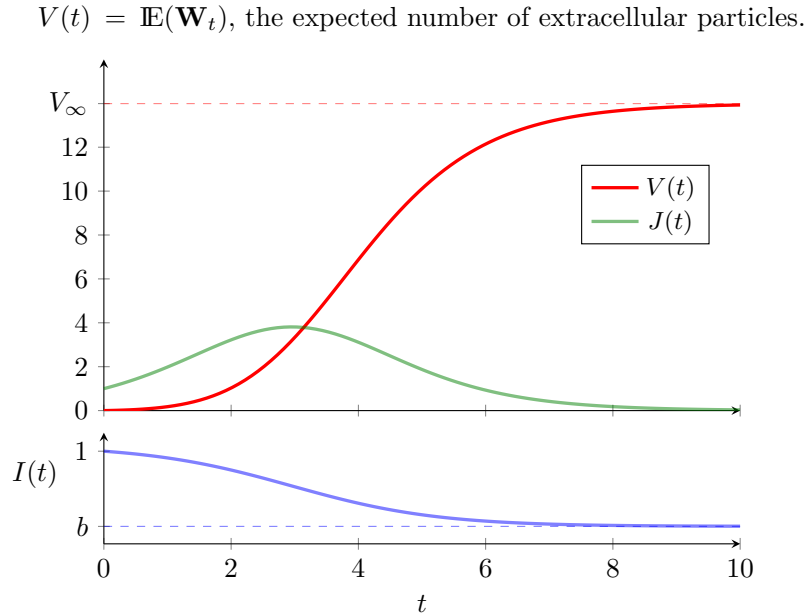


Figure 16: $V(t)$ plotted against time. $\lambda = 1$, $\mu = 0.2$, $\delta = 0.05$.

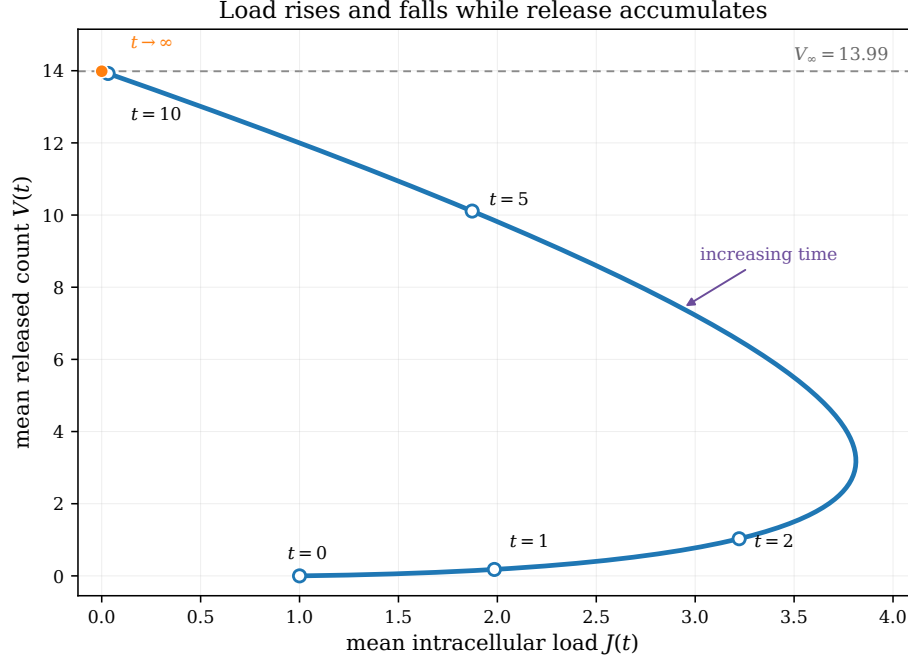


Figure 17: Parametric mean load–release trajectory $(J(t), V(t))$ for $\lambda = 1$, $\mu = 0.2$, and $\delta = 0.05$, with selected times marked along the curve. Starting from $(1, 0)$, the mean load first rises and then falls to zero, while the expected release increases monotonically towards $V_\infty = 13.99$. The joint plot exposes both motions without using time as an axis.

For the case where death is included ($\mu > 0$), V_∞ takes into account eventualities in which the process did not attain catastrophe — the dead realisations contribute nothing to the release. We may extract the mean burst size proper by conditioning on the cases which did experience rupture, effectively discounting the dead realisations. This is done by dividing $V(t)$ by the likelihood of process catastrophe, $1 - I(t)$, following the basic logic

$$\mathbb{E}(\mathbf{W}_t) = \mathbb{E}(\mathbf{W}_t | \text{no rupture}) I(t) + \mathbb{E}(\mathbf{W}_t | \text{rupture}) (1 - I(t)),$$

so that

$$\mathbb{E}(\mathbf{W}_t | \text{rupture}) = \frac{\mathbb{E}(\mathbf{W}_t)}{1 - I(t)}.$$

The ratio $V(t)/[1 - I(t)]$ represents the expected burst size given that catastrophe has taken place by time t . The overall mean burst size, incorporating all realisations which experience rupture regardless of when, is the late-time value of this ratio:

Corollary 4.5. *The mean burst size, conditional on bursting eventually, is*

$$\mathbb{E}(\mathcal{K} | \text{burst}) = \lim_{t \rightarrow \infty} \frac{V(t)}{1 - I(t)} = \frac{V_\infty}{1 - b} = \frac{\lambda - \mu}{\delta} + \frac{1}{1 - b} = \frac{a}{a - 1}, \quad (8)$$

recalling that b is the probability of ultimate process death. Notice the alignment $\mathbb{E}(\mathcal{K} | \text{burst}) = V_\infty$ when $\mu = 0$, since then every realisation bursts.

The full distribution of the burst size $\mathcal{K}$, not only its mean, is derived in Chapter 4a; a preview of its geometric form appears in Section 9.

5 Derivation of equations and results

The closed forms of Section 4 all descend from one Riccati equation, obtained twice over — once for I and once for D — with different initial conditions. We derive that equation, solve it, and then assemble the remaining results from it.

5.1 The Riccati equation for $I(t)$

Considering what may happen during an interval of duration Δt after $t = 0$, we can write down

$$I(0 + \Delta t) = \mu\Delta t + [1 - (\lambda + \mu + \delta)\Delta t]I(0) + \lambda\Delta t I(0)^2 + o(\Delta t).$$

The single initial unit may die, divide, cause rupture, or do nothing at all; each possibility is accounted for on the right-hand side, with the respective event probabilities multiplied by the updated non-rupture likelihoods — as if the process were beginning anew. In the case that a rupture occurred, this value is obviously 0; if the unit dies, non-rupture is certain, contributing the $\mu\Delta t$ term; if a birth takes place, the non-rupture probability assumes the value $I_2(0)$, which, by the branching property, equals $I(0)^2$. Because the model is homogeneous in time, the relation maintains applicability in subsequent intervals $[t, t + \Delta t]$:

$$I(t + \Delta t) = \mu\Delta t + [1 - (\lambda + \mu + \delta)\Delta t]I(t) + \lambda\Delta t I(t)^2 + o(\Delta t).$$

Rearranging yields

$$\frac{I(t + \Delta t) - I(t)}{\Delta t} = \mu - (\lambda + \mu + \delta)I(t) + \lambda I(t)^2 + \frac{o(\Delta t)}{\Delta t},$$

which, in the limit as $\Delta t \rightarrow 0$, becomes

$$\frac{dI}{dt} = \mu - (\lambda + \mu + \delta)I + \lambda I^2, \tag{9}$$

a Riccati equation. Its quadratic right-hand side factorises through the roots a and b defined in Section 2, giving the equivalent form

$$\frac{dI}{dt} = \lambda(I - a)(I - b). \tag{10}$$

Plotting dI/dt against I gives an intuitive picture of why $I(t)$ tends to b at late time: b is the stable root, and any trajectory starting at $I(0) = 1 < a$ runs downhill towards it.

The identities collected in Eq. (1) fall out of this factorisation. Expanding $\lambda(I - a)(I - b)$ and comparing coefficients with (9) gives, by Vieta's formulas,

$$ab = \frac{\mu}{\lambda}, \quad a + b = \frac{\lambda + \mu + \delta}{\lambda}.$$

The two derived constants $A = a - 1$ and $B = 1 - b$ then combine as

$$AB = (a - 1)(1 - b) = a + b - ab - 1 = \frac{\lambda + \mu + \delta}{\lambda} - \frac{\mu}{\lambda} - 1 = \frac{\delta}{\lambda},$$

while $A + B = (a - 1) + (1 - b) = a - b$ and $\theta = \lambda(a - b)$ are immediate from the definitions. These four identities are the bookkeeping that makes every closed form in the chapter collapse to its shortest shape, and they are worth committing to memory.

5.2 Solving the Riccati equation

Since the right-hand side of (10) is a product of linear factors, the equation separates:

$$\int \frac{dI}{(I-a)(I-b)} = \lambda t + \text{const.}$$

The partial-fraction decomposition

$$\frac{1}{(I-a)(I-b)} = \frac{1}{a-b} \left(\frac{1}{I-a} - \frac{1}{I-b} \right)$$

integrates the left-hand side to $(a-b)^{-1} \log \frac{I-a}{I-b}$, so every solution satisfies

$$\frac{I(t)-a}{I(t)-b} = \frac{I(0)-a}{I(0)-b} e^{\lambda(a-b)t} = \frac{I(0)-a}{I(0)-b} w. \quad (11)$$

With $I(0) = 1$, the ratio on the right is $(1-a)/(1-b) = -A/B$, and solving (11) for $I(t)$ reproduces the closed form of Theorem 4.1:

$$I(t) = \frac{aB + bAw}{B + Aw}. \quad (12)$$

(Some treatments of this process label the two roots I_+ and I_- ; the correspondence is $I_+ = a$ and $I_- = b$, and we use a, b throughout.)

5.3 The internal-extinction probability $D(t)$

The probability $D(t) = p_0(t)$ that the process has died out internally by time t satisfies the *same* Riccati equation as I , with a different initial condition. The first-step decomposition is identical in structure: if the first unit dies, the process sits at 0 with certainty; if it divides, both descendant families must die out, an event of probability $D(t)^2$; if it causes a rupture, the process is at H and not at 0. Hence

$$\frac{dD}{dt} = \mu - (\lambda + \mu + \delta)D + \lambda D^2, \quad D(0) = 0. \quad (13)$$

In the ratio form (11), the initial ratio is now $(0-a)/(0-b) = a/b$, so

$$\frac{D(t)-a}{D(t)-b} = \frac{a}{b} w,$$

and solving for $D(t)$ gives the closed form of Theorem 4.2:

$$D(t) = \frac{ab(w-1)}{aw-b}.$$

That I and D obey the same differential equation is not a coincidence: both are “survival-type” quantities for the same process, distinguished only by which absorbing state is counted as survival, and the shared equation is what makes their difference simplify so cleanly.

5.4 The non-fixation probability $I_{\text{fix}}(t)$

With I and D in hand, $I_{\text{fix}} = I - D$ could be written down immediately; but the direct derivation is worth recording, because it exposes the structure that the multi-founder formulas of Section 4 inherit. A first-step decomposition for I_{fix} differs from the two above in the birth branch: after

a division, non-fixation of the pair requires that *neither* family fixates, whether by death or by rupture, and that event has probability $I^2 - D^2$, not I_{fix}^2 . Thus

$$\frac{dI_{\text{fix}}}{dt} = -(\lambda + \mu + \delta)I_{\text{fix}} + \lambda(I^2 - D^2), \quad I_{\text{fix}}(0) = 1, \quad (14)$$

where the death and catastrophe branches contribute nothing, each sending the process to one of the absorbing states. Subtracting (13) from (9) yields the same equation, confirming $I_{\text{fix}} = I - D$. The closed form follows by algebra:

$$I_{\text{fix}}(t) = I(t) - D(t) = \frac{aB + bAw}{B + Aw} - \frac{ab(w-1)}{aw - b} = \frac{(a-b)^2 w}{(B + Aw)(aw - b)}.$$

The limits read off from this expression agree with the definition: $I_{\text{fix}}(0) = (a-b)^2 / [(A+B)(a-b)] = 1$, and $I_{\text{fix}}(t) \rightarrow 0$ as $w \rightarrow \infty$, since the numerator grows like w while the denominator grows like w^2 . Differentiating (14) at $t = 0$ gives the initial slope $I'_{\text{fix}}(0) = -(\lambda + \mu + \delta) + \lambda = -(\mu + \delta)$, the total rate at which a productive realisation is lost at the first instant. When $\mu = 0$ we have $b = 0$ and $D \equiv 0$, and (14) collapses to the Riccati equation for I itself.

5.5 The forward relation $I' = -\delta J$

One further relation, connecting the non-catastrophe probability to the mean load, will be used repeatedly. The forward Kolmogorov equation for the probability of the rupture state is

$$\frac{d}{dt} \Pr\{\mathbf{X}_t = H\} = \delta \sum_{n \geq 1} n p_n(t) = \delta J(t) :$$

the state H is entered from every productive state n , at rate δn . Since $I(t) = 1 - \Pr\{\mathbf{X}_t = H\}$, this gives immediately

$$\frac{dI}{dt} = -\delta J, \quad (15)$$

which is the first of the key ODEs of Section 2 in a second guise, and from which $J(t) = -\delta^{-1} I'(t)$ follows by differentiation of (12), as claimed in Theorem 4.3.

6 Moment equations for $J(t)$ and $V(t)$

The closed forms of Theorems 4.3 and 4.4 were stated in Section 4, and the relation $I' = -\delta J$ that connects them was derived in Section 5.5. Here we derive the remaining moment equations, which are obtained by tracking how the expectations change over a short interval Δt .

6.1 Rate of change of J

Individual realisations of the stochastic process $\mathbf{X}_t$ vary greatly. When comparing a large number side by side (assuming $\lambda > \mu$), some counts will have collapsed to 0 while those remaining climb higher and higher. Despite this, we can quantify the expected value of the process over time — call it $J(t) = \mathbb{E}(\mathbf{X}_t \mid \mathbf{X}_0 = 1)$.

By looking to the list of possible events over a period Δt , we can write down the relationship describing how $\mathbb{E}(\mathbf{X}_t)$ changes:

$$\mathbb{E}(\mathbf{X}_{t+\Delta t}) = \mathbb{E}(\mathbf{X}_t) + \mathbb{E}(\mathbf{X}_{t+\Delta t} - \mathbf{X}_t);$$

the expected value of $\mathbf{X}_{t+\Delta t}$ is what it was at time t , plus the expected change during the interval $[t, t + \Delta t]$.

$$\mathbb{E}(\mathbf{X}_{t+\Delta t}) = \mathbb{E}(\mathbf{X}_t) + \mathbb{E}(\lambda \Delta t \mathbf{X}_t - \mu \Delta t \mathbf{X}_t - \delta \Delta t \mathbf{X}_t \cdot \mathbf{X}_t). \quad (16)$$

The final component, $-\delta \Delta t \mathbf{X}_t \cdot \mathbf{X}_t$, represents the probability that a rupture occurs during the interval, $\delta \Delta t \mathbf{X}_t$, multiplied by the change to $\mathbf{X}_t$ — in this case $-\mathbf{X}_t$, the complete elimination of the intracellular population. Rearranging (16), we find

$$\frac{\mathbb{E}(\mathbf{X}_{t+\Delta t}) - \mathbb{E}(\mathbf{X}_t)}{\Delta t} = \lambda \mathbb{E}(\mathbf{X}_t) - \mu \mathbb{E}(\mathbf{X}_t) - \delta \mathbb{E}(\mathbf{X}_t^2),$$

which, evaluated in the limit $\Delta t \rightarrow 0$, becomes

$$\frac{dJ}{dt} = (\lambda - \mu)J - \delta K(t), \quad (17)$$

with $K(t) = \mathbb{E}(\mathbf{X}_t^2)$.

6.2 Rate of change of V , and its solution

Recall from Section 3 the secondary process $\mathbf{W}_t$, the number of units expelled following rupture: $\mathbf{W}_t$ remains at 0 unless the associated process $\mathbf{X}_t$ undergoes a catastrophe, and with probability $1 - b$ this happens eventually, in which case $\mathbf{W}_t$ jumps from 0 to the value of $\mathbf{X}_t$ immediately prior to expulsion. By reasoning analogous to that which derived (17),

$$\frac{dV}{dt} = \delta K(t), \quad (18)$$

where $V(t) = \mathbb{E}(\mathbf{W}_t)$: the expected released count increases exactly when rupture occurs, and then by the pre-rupture load.

Equations (17) and (18) combine to give

$$\frac{dJ}{dt} = (\lambda - \mu)J - \frac{dV}{dt},$$

which, with (15), rearranges into

$$\frac{dV}{dt} = -\frac{(\lambda - \mu)}{\delta} \frac{dI}{dt} - \frac{dJ}{dt}.$$

Then, integrating,

$$V(t) = -\frac{(\lambda - \mu)}{\delta} I(t) - J(t) + \text{const},$$

and finally, with initial conditions $V(0) = 0$ and $J(0) = I(0) = 1$,

$$V(t) = \frac{(\lambda - \mu)}{\delta} [1 - I(t)] - J(t) + 1, \quad (19)$$

which is the first form stated in Theorem 4.4; the equivalent form $(1 - I)[1 + \frac{\lambda}{\delta}(1 - I)]$ follows on replacing I' with the Riccati right-hand side of (9) and collecting terms.

In late time, $V(t)$ tends to

$$V_\infty = \frac{(\lambda - \mu)}{\delta} [1 - b] + 1 = \frac{aB}{A},$$

as $I \rightarrow b$ and $J \rightarrow 0$, which can be interpreted as the expected rupture size counting all realisations, the dead ones included with a release of zero. When $\mu = 0$, every realisation ruptures and

$$V_\infty = 1 + \frac{\lambda}{\delta}.$$

7 Variance and standard deviation of $\mathbf{X}_t$ and $\mathbf{W}_t$

To describe the variance of a random variable, we use the standard formula

$$\text{Var}(\mathbf{X}_t) = \mathbb{E}(\mathbf{X}_t^2) - \mathbb{E}(\mathbf{X}_t)^2,$$

and we apply it to $\mathbf{X}_t$ and $\mathbf{W}_t$ to quantify the variability of intracellular and extracellular particle counts as functions of time. In each case, we first require a formula for the expected value of the count squared: $\mathbb{E}(\mathbf{X}_t^2)$ and $\mathbb{E}(\mathbf{W}_t^2)$.

7.1 Variance of $\mathbf{X}_t$

7.1.1 Formula for $\mathbb{E}(\mathbf{X}_t^2)$ ($= K(t)$)

A formula for $K(t) = \mathbb{E}(\mathbf{X}_t^2)$ is derived in the following way. Take the time derivative of the two equations

$$(15): \quad dI/dt = -\delta J, \quad (9): \quad dI/dt = \mu - (\lambda + \mu + \delta)I + \lambda I^2.$$

From the first,

$$\frac{dJ}{dt} = -\frac{1}{\delta} \frac{d^2 I}{dt^2}, \quad (20)$$

while differentiating the second gives

$$\frac{d^2 I}{dt^2} = -(\lambda + \mu + \delta) \frac{dI}{dt} + 2\lambda I \frac{dI}{dt},$$

which, with $dI/dt = -\delta J$, becomes

$$\frac{d^2 I}{dt^2} = (\lambda + \mu + \delta)\lambda J - 2\lambda \delta I J. \quad (21)$$

Comparing (21) with (20),

$$\frac{dJ}{dt} = -(\lambda + \mu + \delta)J + 2\lambda I J;$$

then, equating this with (17),

$$(\lambda - \mu)J - \delta K = -(\lambda + \mu + \delta)J + 2\lambda I J.$$

Rearranging for $K(t)$,

$$K(t) = \frac{2\lambda}{\delta} \left(1 + \frac{\delta}{2\lambda} - I\right) J = \left[1 + \frac{2\lambda}{\delta}(1 - I)\right] J = \frac{2\lambda}{\delta}(\kappa - I)J, \quad (22)$$

with

$$\kappa = 1 + \frac{\delta}{2\lambda}.$$

Theorem 7.1. *The second moment of the process state is*

$$K(t) = \left[1 + \frac{2\lambda}{\delta}(1 - I)\right] J = \frac{2\lambda}{\delta}(\kappa - I)J, \quad (23)$$

and, written as a function of I alone using $J = -\delta^{-1}I' = -\frac{\lambda}{\delta}(I - a)(I - b)$,

$$K(I) = \frac{2\lambda^2}{\delta^2}(I - \kappa)(I - a)(I - b). \quad (24)$$

The factored form (24) shows at a glance that K shares the vanishing of J at both absorbing limits $I \rightarrow a$ and $I \rightarrow b$, as it must: a fixed process has no second moment to speak of either.

7.1.2 One standard deviation from the mean, $J(t)$

With the expectation of $\mathbf{X}_t^2$ in the form (22), we can write down the variance of $\mathbf{X}_t$ using $\text{Var}(\mathbf{X}_t) = \mathbb{E}(\mathbf{X}_t^2) - \mathbb{E}(\mathbf{X}_t)^2$:

$$\begin{aligned}\text{Var}(\mathbf{X}_t) &= K(t) - J(t)^2 \\ &= \frac{2\lambda}{\delta} (\kappa - I) J - J^2,\end{aligned}$$

and the standard deviation is then

$$\text{Std}(\mathbf{X}_t) = \sqrt{\frac{2\lambda}{\delta} (\kappa - I) J - J^2}.$$

7.2 Variance of $\mathbf{W}_t$

We can likewise compute the variance of $\mathbf{W}_t$ using $\text{Var}(\mathbf{W}_t) = \mathbb{E}(\mathbf{W}_t^2) - \mathbb{E}(\mathbf{W}_t)^2$. To begin, we must find a formula for $\mathbb{E}(\mathbf{W}_t^2)$.

7.2.1 Formula for $\mathbb{E}(\mathbf{W}_t^2)$

Accounting for changes to $\mathbf{W}_t^2$ over an infinitesimal time Δt ,

$$\mathbb{E}(\mathbf{W}_{t+\Delta t}^2) = \mathbb{E}(\mathbf{W}_t^2) + \mathbb{E}(\mathbf{W}_{t+\Delta t}^2 - \mathbf{W}_t^2),$$

we can derive the equation

$$\frac{d\mathbb{E}(\mathbf{W}_t^2)}{dt} = \delta \mathbb{E}(\mathbf{X}_t^3) : \quad (25)$$

when rupture occurs from state n , at rate δn , the released count jumps from 0 to n , so $\mathbf{W}_t^2$ gains n^2 , and the rate-times-jump contribution is δn^3 . This might seem like a dead end, since we have yet to produce a description of $\mathbb{E}(\mathbf{X}_t^3)$. As it happens, the problem can be sidestepped by thinking about the rate of change of $K(t) = \mathbb{E}(\mathbf{X}_t^2)$:

$$\mathbb{E}(\mathbf{X}_{t+\Delta t}^2) = \mathbb{E}(\mathbf{X}_t^2) + \mathbb{E}(\mathbf{X}_{t+\Delta t}^2 - \mathbf{X}_t^2). \quad (26)$$

Because of the quadratic, changes in $\mathbf{X}_t^2$ require special attention (Table 1).

Probability of occurrence	Change in $\mathbf{X}_t$	Change in $\mathbf{X}_t^2$
$\lambda \mathbf{X}_t \Delta t$	1	$(\mathbf{X}_t + 1)^2 - \mathbf{X}_t^2 = 2\mathbf{X}_t + 1$
$\mu \mathbf{X}_t \Delta t$	-1	$(\mathbf{X}_t - 1)^2 - \mathbf{X}_t^2 = -2\mathbf{X}_t + 1$
$\delta \mathbf{X}_t \Delta t$	$-\mathbf{X}_t$	$(\mathbf{X}_t - \mathbf{X}_t)^2 - \mathbf{X}_t^2 = -\mathbf{X}_t^2$

Table 1: The three event types and their contributions to $\mathbf{X}_t^2$.

From the possibilities in the table,

$$\mathbb{E}(\mathbf{X}_{t+\Delta t}^2 - \mathbf{X}_t^2) = \mathbb{E}(\lambda \mathbf{X}_t \Delta t \cdot (2\mathbf{X}_t + 1) + \mu \mathbf{X}_t \Delta t \cdot (-2\mathbf{X}_t + 1) + \delta \mathbf{X}_t \Delta t \cdot (-\mathbf{X}_t^2))$$

$$= \left[2(\lambda - \mu)\mathbb{E}(\mathbf{X}_t^2) + (\lambda + \mu)\mathbb{E}(\mathbf{X}_t) - \delta\mathbb{E}(\mathbf{X}_t^3) \right] \Delta t.$$

Taking this with (26), the following equation is obtained in the $\Delta t \rightarrow 0$ limit:

$$\frac{dK}{dt} = 2(\lambda - \mu)K + (\lambda + \mu)J - \delta\mathbb{E}(\mathbf{X}_t^3). \quad (27)$$

Equations (25) and (27) together eliminate the third moment:

$$\frac{d\mathbb{E}(\mathbf{W}_t^2)}{dt} = 2(\lambda - \mu)K + (\lambda + \mu)J - \frac{dK}{dt}. \quad (28)$$

Integrating from 0 to t , and using $V' = \delta K$ (18) and $I' = -\delta J$ (15) to evaluate the two integrals,

$$\begin{aligned} \mathbb{E}(\mathbf{W}_t^2) &= 2(\lambda - \mu) \int_0^t K \, ds + (\lambda + \mu) \int_0^t J \, ds - K(t) + K(0) \\ &= \frac{2(\lambda - \mu)}{\delta} V(t) + \frac{\lambda + \mu}{\delta} (1 - I(t)) - K(t) + K(0). \end{aligned}$$

The integration constant here is $K(0) = \mathbb{E}(\mathbf{X}_0^2) = 1$, since the process begins with a single unit, and $\mathbb{E}(\mathbf{W}_0^2) = 0$ contributes nothing on the left. Therefore:

Theorem 7.2. *The second moment of the released count is*

$$\mathbb{E}(\mathbf{W}_t^2) = \frac{2(\lambda - \mu)}{\delta} V - \frac{\lambda + \mu}{\delta} I - K + \frac{\lambda + \mu}{\delta} + 1. \quad (29)$$

As a check, when $\mu = 0$ we have $b = 0$, so every realisation ruptures, $I(\infty) = 0$ and $K(\infty) = 0$, while $V(\infty) = 1 + \lambda/\delta$; the formula gives

$$\mathbb{E}(\mathbf{W}_t^2) \Big|_{t \rightarrow \infty} = \frac{2\lambda}{\delta} \left(1 + \frac{\lambda}{\delta}\right) + \frac{\lambda}{\delta} + 1 = \frac{2\lambda^2}{\delta^2} + \frac{3\lambda}{\delta} + 1.$$

This is exactly the second moment of the geometric burst-size law $P_k = (\delta/\lambda)a^{-k}$ with $a = (\lambda + \delta)/\lambda$ — a geometric distribution on $\{1, 2, \dots\}$ with success probability $\delta/(\lambda + \delta)$, whose second moment is $(2 - s)/s^2$ for $s = \delta/(\lambda + \delta)$. The burst-size law itself is derived in Chapter 4a; for now the agreement confirms (29).

7.2.2 One standard deviation from the mean, $V(t)$

Using (29), it is possible to write down a formula for the variance of $\mathbf{W}_t$:

$$\text{Var}(\mathbf{W}_t) = \frac{2(\lambda - \mu)}{\delta} V - \frac{\lambda + \mu}{\delta} I - K + \frac{\lambda + \mu}{\delta} + 1 - V^2. \quad (30)$$

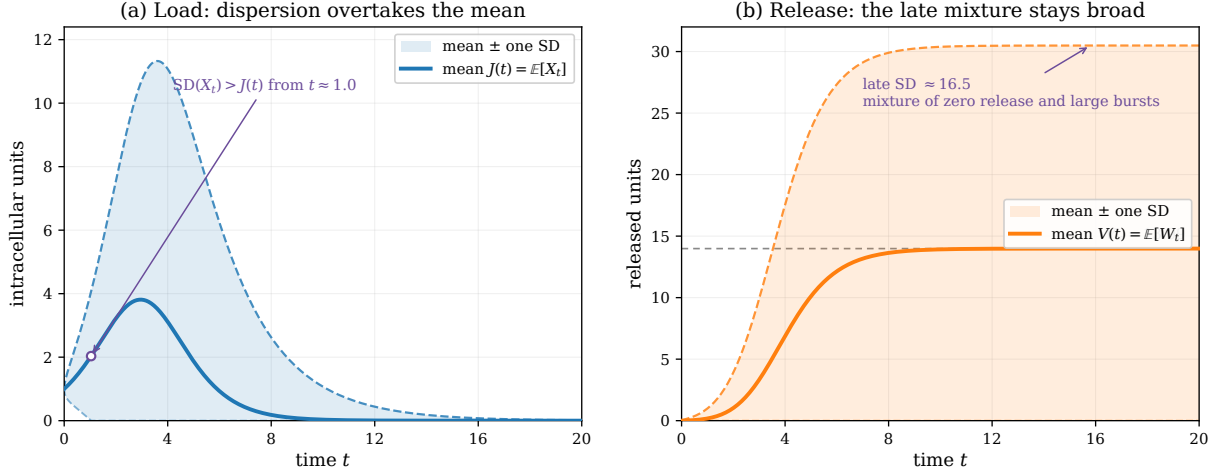


Figure 18: Mean and one-standard-deviation envelopes for the intracellular load $\mathbf{X}_t$ (a) and released count $\mathbf{W}_t$ (b), for $\lambda = 1$, $\mu = 0.2$, and $\delta = 0.05$. The lower envelope is clipped at zero, the support boundary for both counts. Load dispersion overtakes $J(t)$ early in the productive window, while the late spread in $\mathbf{W}_t$ records the mixture of never-rupturing paths and large releases. Means alone therefore conceal substantial pathwise uncertainty.

8 Variable dependencies on $I(t)$

It happens to be possible to write all of the formulas $J(t)$, $\text{Var}(\mathbf{X}_t)$, $V(t)$ and $\text{Var}(\mathbf{W}_t)$ as functions of the non-rupture probability $I(t)$ alone. Time enters these expressions only through I , so the whole family of curves is parametrised by a single quantity moving monotonically from 1 down to b ; this is the viewpoint adopted in Chapter 4b, where the single-cell formulas become the kernels of a population model.

8.1 $J(I)$ and $K(I)$

Since $J = -\delta^{-1} dI/dt$ (15), and $dI/dt = \lambda(I - a)(I - b)$ (10), we can write J in terms of I :

$$J(I) = -\frac{\lambda}{\delta}(I - a)(I - b). \quad (31)$$

Combining this with Theorem 7.1, the second moment likewise depends on I alone:

$$K(I) = \frac{2\lambda^2}{\delta^2}(I - \kappa)(I - a)(I - b), \quad (32)$$

with $\kappa = 1 + \delta/2\lambda$.

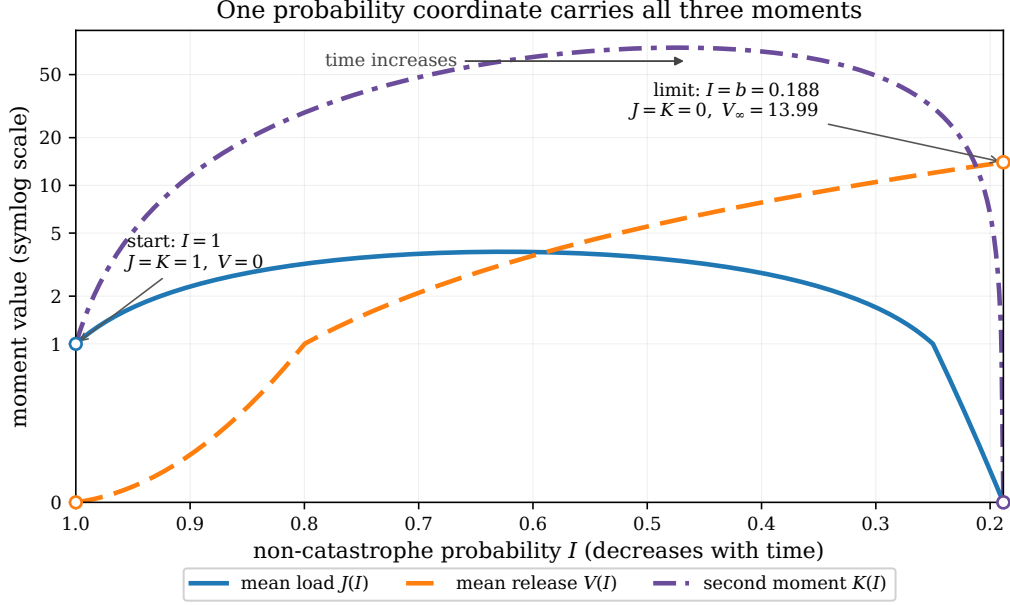


Figure 19: The I -economy of the mean load J , mean release V , and second moment K for $\lambda = 1$, $\mu = 0.2$, and $\delta = 0.05$. Time runs from left to right as I decreases from 1 to b ; the symmetric-log ordinate retains the raw moment values while keeping all three curves legible. Their labelled endpoints emphasise that time merely reparametrises a family of quantities already fixed algebraically by I .

8.2 Mean and variance of $\mathbf{X}_t$

The variance of $\mathbf{X}_t$ follows from $\text{Var}(\mathbf{X}_t) = K(I) - J(I)^2$ written entirely in terms of I :

$$\begin{aligned} \text{Var}(\mathbf{X}_t) &= K(I) - J(I)^2 \\ &= \frac{2\lambda^2}{\delta^2} (I - \kappa) (I - a) (I - b) - \left(\frac{\lambda}{\delta} (I - a)(I - b) \right)^2, \end{aligned}$$

which, like J and K , vanishes at either absorbing limit $I \rightarrow a$ or $I \rightarrow b$ and is positive between them.

8.3 Mean and variance of $\mathbf{W}_t$

As we have defined $V(t)$ in terms of $I(t)$ and $J(t)$, it is likewise possible to use (31) to write V as a function only of the non-rupture probability:

$$V(I) = \frac{\lambda - \mu}{\delta} (1 - I) - J(I) + 1 = \frac{\lambda - \mu}{\delta} (1 - I) + \frac{\lambda}{\delta} (I - a)(I - b) + 1. \quad (33)$$

And given (30), we can do the same for the variance of $\mathbf{W}_t$:

$$\text{Var}(\mathbf{W}_t) = \frac{2(\lambda - \mu)}{\delta} V(I) - \frac{\lambda + \mu}{\delta} I - K(I) + \frac{\lambda + \mu}{\delta} + 1 - V(I)^2. \quad (34)$$

Written out in full, with (31), (32) and (33) substituted:

$$\begin{aligned} \text{Var}(\mathbf{W}_t) = & \frac{2(\lambda - \mu)}{\delta} \left[\frac{\lambda - \mu}{\delta} (1 - I) + \frac{\lambda}{\delta} (I - a)(I - b) + 1 \right] \\ & - \frac{\lambda + \mu}{\delta} I - \frac{2\lambda^2}{\delta^2} (I - \kappa)(I - a)(I - b) + \frac{\lambda + \mu}{\delta} + 1 \\ & - \left[\frac{\lambda - \mu}{\delta} (1 - I) + \frac{\lambda}{\delta} (I - a)(I - b) + 1 \right]^2. \end{aligned}$$

The late-time value of (34) is the variance of the burst size (over all realisations, dead ones contributing zero); at $I = b$ with $\mu > 0$ this remains finite, and when $\mu = 0$ it reduces to the variance of the geometric law of Section 7, namely $(1 - s)/s^2$ with $s = \delta/(\lambda + \delta)$.

9 What comes next

The results of this chapter describe the BDC process through its first and second moments: the non-catastrophe and non-fixation probabilities, the mean load, the mean release, and their variances, each in closed form and each written as a function of the single quantity $I(t)$. Yet several of the formulas we have derived hint at a richer structure that moments alone do not capture. The ratio $V(t)/[1 - I(t)]$ conditioned on rupture; the geometric form of the burst size that surfaced in the $\mu = 0$ check of Theorem 7.2; the finite plateau of the conditional mean J/I in Figure 14's neighbourhood — each of these points beyond the expectations, towards the full distributions of the process. Completing that picture, and then embedding the single cell in a population, are the two directions in which this thesis now moves.

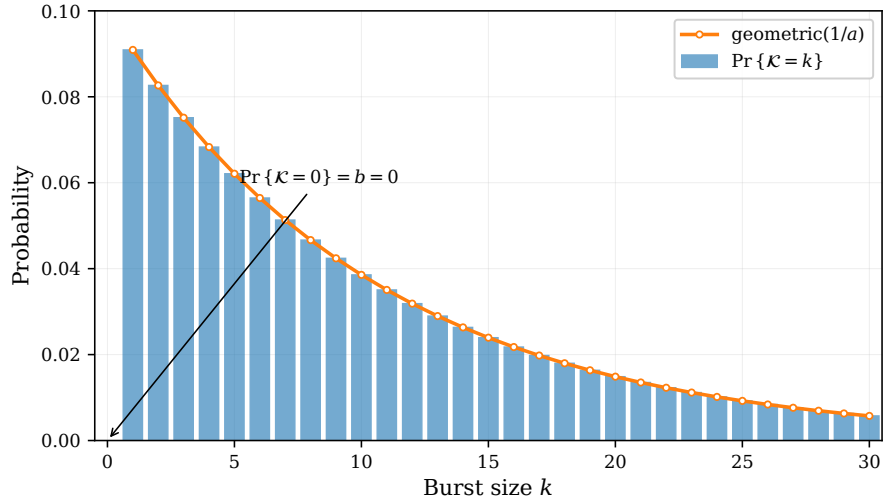


Figure 20: Burst-size preview for $\mu = 0$: $\Pr\{\mathcal{K} = k\} = (\delta/\lambda)a^{-k}$ coincides with $\text{geometric}(1/a)$, while the mass at $\mathcal{K} = 0$ is $b = 0$. Parameters $\lambda = 1$, $\mu = 0$, $\delta = 0.1$, and $a = 1.1$.

Completing the single-cell theory

Chapter 4a completes the description of one infected cell. It solves the probability generating function of the process and thereby obtains the state probabilities $p_n(t)$ at every time — not merely their sums I , D and I_{fix} — and with them the quasi-stationary distribution: the limiting law of the intracellular load conditioned on the cell remaining productive. The burst time and burst size

receive their full treatment there: the defective density of τ , the exact burst-size law $P_k = (\delta/\lambda)a^{-k}$ whose $\mu = 0$ preview appears above, and the conditional laws of burst size given burst time and of burst time given burst size. The chapter also takes up the multi-founder formulas of Section 4, which become essential when a cell is infected by several particles at once, and it ends with the chained immediate-transfer model, in which the released particles of one cell found the next infection without delay. Everything in Chapter 4a rests on the closed forms and identities of the present chapter, and nothing beyond them.

Embedding the cell in a population

Chapter 4b turns from one cell to many. A released burst does not end the story: the particles may go on to infect fresh cells, each starting its own BDC trajectory. Modelling that population faithfully requires the single-cell quantities of this chapter as *age-dependent kernels* — the survival kernel $I_{\text{fix}}(a)$, which says how many of the cells infected a time units ago are still productive, and the release kernel $g(a) = \delta K(a)$, which says when and how much they release. With these kernels the population dynamics become a renewal system, and the constant-release ordinary differential equations conventionally used in viral dynamics emerge only as an approximation — one that Chapter 4b shows to be quantitatively wrong in the regimes that matter for early infection. The comparison between bursting and its alternative, budding release, is taken up there as well.

Two open questions of the present chapter are answered by these developments: the origin of the geometric burst-size law glimpsed in Theorem 7.2’s check (Chapter 4a derives it from the state probabilities), and the status of the conditional plateau of J/I (it is the mean of the quasi-stationary distribution, and the burst size turns out to be distributed exactly as that quasi-stationary law). What this chapter cannot settle — the evolutionary implications of releasing in one large burst rather than continuously, and the behaviour of the population when released particles compete for target cells — is where the later chapters begin.

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